

# **Hospital Operations Analytics**

## **Exploratory Data Analysis (EDA) Report**

### **>>> PREPARED BY:**

*- AKSHITA JOSHI,*

*JUNIOR DATA ANALYST INTERN.*

### **>>> TOOLS USED:**

*> SQLite*

*> Python*

*> Pandas*

*> NumPy*

*> Jupyter Notebook*

*> Microsoft Word*

### **>>> PROJECT:**

*- Hospital Operations Analytics Internship.*

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## **1.EXECUTIVE SUMMARY-**

This Exploratory Data Analysis (EDA) was conducted on the Hospital Operations Analytics database to evaluate the structure, quality, and completeness of the hospital's operational data before performing advanced analytics and dashboard development. The analysis was carried out using Python, SQLite, Pandas, and NumPy.

The database consists of eight interconnected tables representing different operational domains, including patients, doctors, departments, appointments, encounters, billing, payments, and prescriptions. Data quality assessment confirmed that the dataset is highly reliable, with no duplicate records across any table and very few missing values. Descriptive statistics and categorical frequency analysis were performed to understand the characteristics of each dataset and identify preliminary business insights.

Overall, the database is well-structured and suitable for SQL analysis, KPI development, business intelligence reporting, and Power BI dashboard creation.

## **2. PROJECT OBJECTIVE-**

-The objective of this exploratory data analysis is to understand the structure, quality, and characteristics of the Hospital Operations Analytics database.

-The analysis focuses on validating data quality, generating descriptive statistics, examining categorical distributions, and identifying initial operational insights.

-These findings establish a strong analytical foundation for subsequent SQL querying, KPI generation, and dashboard development.

### 3. DATASET OVERVIEW-

- Based on the EDA, the hospital database contains eight relational tables.

| TABLE                       | RECORDS | PURPOSE                |
|-----------------------------|---------|------------------------|
| <b><i>Patients</i></b>      | 30      | Patient Demographics   |
| <b><i>Doctors</i></b>       | 10      | Doctor Information     |
| <b><i>Departments</i></b>   | 5       | Hospital Departments   |
| <b><i>Appointments</i></b>  | 50      | Appointment Scheduling |
| <b><i>Encounters</i></b>    | 35      | Clinical Consultations |
| <b><i>Billing</i></b>       | 35      | Billing Information    |
| <b><i>Payments</i></b>      | 35      | Payment Transactions   |
| <b><i>Prescriptions</i></b> | 60      | Medication Records     |

### 3. **DATA QUALITY ASSESSMENT-**

--- Overall, the data quality is excellent.

#### **Strengths-**

- No duplicate records were found in any table.
- Most datasets contain complete information with no missing values.
- Consistent data types are maintained across identifiers, numerical values, dates, and categorical attributes.
- The database follows a clear relational structure that supports efficient joins and analytical queries.

#### **Observation-**

- The only notable issue is within the **Departments** table, where the head\_of\_department field is empty for all five records. This field should ideally be populated in future data collection to support organizational reporting.

## **4. TABLE-WISE ANALYSIS-**

### **4.1 Patients:**

The Patients table contains **30 patient records** with **10 attributes**, including demographic details, blood group, contact information, and registration date. No missing values or duplicate records were identified, indicating excellent data quality. Gender distribution is perfectly balanced, with **15 male** and **15 female** patients. All registered patients currently have the **O+ blood group**, suggesting that the dataset is sample data rather than a real-world demographic distribution. This table forms the foundation for linking appointments, encounters, billing, and prescriptions.

### **4.2 Doctors:**

The Doctors table consists of **10 doctors** distributed across **five medical specializations**. The average professional experience is **8.5 years**, with experience ranging from **4 to 13 years**, indicating a balanced workforce of mid-level and experienced clinicians. Emergency Medicine has the highest representation with **three doctors**, while Cardiology, Neurology, and Orthopedics each have two specialists and Pediatrics has one. No missing values or duplicate records were observed, making this dataset reliable for workforce and staffing analysis.

### 4.3 DEPARTMENTS:

The Departments table contains **five hospital departments**; each located in a different area of the hospital. Every department appears exactly once, indicating a clean master reference table. However, the head\_of\_department field is missing for all records, representing the only significant data completeness issue identified during the EDA. Apart from this column, the dataset is complete and suitable for department-level analysis.

### 4.4 APPOINTMENTS:

The Appointments table contains **50 appointment records**. Of these, **35 appointments (70%)** have been completed, while the remaining records are evenly distributed among Scheduled, Cancelled, and No Show statuses (five records each). This indicates a high appointment completion rate and provides useful information for analysing operational efficiency, patient attendance, and doctor workload.

### 4.5 ENCOUNTERS:

The Encounters table records **35 completed clinical consultations** and provides a reliable basis for analysing consultation activity and clinical workload. Consultation duration ranges from **15 to 29 minutes**, with an average duration of approximately **21.4 minutes**. This narrow duration range suggests a consistent consultation process across patient visits. No missing values or duplicate records were found,



making the dataset suitable for clinical workload and operational analysis.

## 4.6 Billing:

The Billing table contains **35 billing records** with complete financial information. The average bill amount is **₹4,000**, with billing values ranging from **₹2,000** to **₹6,000**. The moderate variation in billing amounts indicates differences in treatment complexity while maintaining relatively consistent pricing across services. This table forms the basis for revenue and financial performance analysis.

## 4.7 Payments:

The Payments table consists of **35 payment transactions**. The average payment amount is approximately **₹3,828.57**, while the median payment equals **₹4,000**. Payment methods are almost equally distributed across Cash, Card, Insurance, and UPI, indicating multiple payment options for patients. Importantly, **32 payments have been completed**, while only **3 remain pending**, demonstrating a high payment completion rate and effective revenue collection.

## 4.8 Prescriptions:

The Prescriptions table contains **60 prescription records** linked to patient encounters. Every medication appears only once in the sample dataset, and all prescriptions have a standardized treatment duration of **7 days**. No missing values or duplicate records were identified. Although this synthetic dataset does not reveal common prescribing patterns, it provides a structured foundation for medication analytics and prescription monitoring.

## **5. KEY FINDINGS-**

- The database demonstrates **excellent overall data quality**, with no duplicate records across any of the eight tables.
- The only significant data quality issue is the missing head\_of\_department information in the Departments table.
- The hospital maintains a balanced medical workforce, with doctors averaging **8.5 years** of experience.
- **70% of appointments** have been successfully completed, indicating efficient appointment management.
- Clinical consultations average approximately **21 minutes**, reflecting a consistent consultation workflow.
- Average billing per encounter is **₹4,000**, while the average payment received is approximately **₹3,828.57**.
- More than **91% of payments** have been completed, suggesting strong financial collection performance.

## **6. RECOMMENDATIONS-**

- Populate the head\_of\_department field for all departments to improve organizational reporting.
- Expand demographic diversity in the patient dataset for more representative healthcare analytics.
- Monitor cancelled and no-show appointments to improve scheduling efficiency.
- Track pending payments regularly to further strengthen revenue collection.
- Use the current dataset as the foundation for KPI development and Power BI dashboard creation.

## **7. CONCLUSION-**

-The exploratory data analysis confirms that the Hospital Operations Analytics database is clean, well-structured, and suitable for advanced analytics.

-The combination of high-quality data, normalized relationships, and complete operational records provides a strong foundation for SQL analysis, KPI generation, business intelligence reporting, and interactive dashboard development.

-The insights generated during this phase will support the next stages of the project, including business query analysis and Power BI visualization.